

PAPER_279 — Sombrero SMBH Dominance Ratio γ_{BH} and UQFF Sphere of Influence r_{SOI}

Author: Daniel T. Murphy **Module:** SOMBRERO_UQFF_MODULE.cpp (UQFF 2.0) **Session:** 77 — March 2026 **Framework:** Unified Quantum Field Framework (UQFF 2.0) **Status:** Complete — embedded in SOMBRERO_UQFF_MODULE.cpp **Whitepaper Series:** 279/1000 **DOI (Provisional):** UQFF-2026-279-BH

Abstract

The Sombrero Galaxy (M104) harbours one of the most massive black holes relative to its host galaxy mass of any object in the Local Universe: $M_{\text{BH}} = 10^9 M_{\text{sun}}$ in a galaxy of total mass $M = 10^{11} M_{\text{sun}}$, giving a **SMBH Dominance Ratio** $\gamma_{\text{BH}} = M_{\text{BH}}/M = 0.01$ (1%). For comparison, the Milky Way's Sgr A* has $\gamma_{\text{BH}} \approx 4 \times 10^{-5}$ (~0.004%); Sombrero's SMBH is **250× more dominant relative to its host**. Within the UQFF framework, we define the **UQFF Sphere of Influence** $r_{\text{SOI}} = r \times \sqrt{(\gamma_{\text{BH}})}$, the radius at which the central BH's direct gravitational contribution equals the total galaxy contribution at the reference radius. For Sombrero, $r_{\text{SOI}} = 2.36 \times 10^{19} \text{ m}$ — a precise UQFF prediction setting the boundary inside which BH gravity exceeds galaxy-mean gravity in the UQFF model.

UQFF Discovery: Novel application of UQFF calibration constants ($\tau = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[\text{SSq}] = 0.57$) uniquely enabling this analysis establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Observational Motivation

1.1 The Sombrero SMBH

The Sombrero Galaxy's central black hole mass has been measured through stellar and gas kinematics:

- Ford et al. (1996): $M_{\text{BH}} = (1.0 \pm 0.75) \times 10^9 M_{\text{sun}}$ (gas kinematics, HST)
- Kormendy et al. (1996): confirmation from stellar dynamics
- Jardel et al. (2011): $M_{\text{BH}} \approx 6.6 \times 10^8 M_{\text{sun}}$ (JAM modelling, consistent with $\sim 10^9$ range)

We adopt $M_{\text{BH}} = 1.0 \times 10^9 M_{\text{sun}} = 1.989 \times 10^{39} \text{ kg}$.

The total galaxy mass (stars + gas + DM within the reference radius) is $M = 10^{11} M_{\text{sun}} = 1.989 \times 10^{41} \text{ kg}$.

1.2 Why γ_{BH} Matters

In standard astrophysical models, BH sphere-of-influence calculations use the velocity dispersion ($r_{\text{SOI}} = GM_{\text{BH}}/\sigma^2$). The UQFF framework generalises this to a mass-ratio-based definition consistent with the 26-layer Triadic gravity structure, yielding a dimensionless parameter γ_{BH} that naturally encodes the BH's dominance within the UQFF vacuum field hierarchy.

2. Theoretical Derivation

2.1 SMBH Dominance Ratio

We define the dimensionless **SMBH Dominance Ratio**:

$$\gamma_{\text{BH}} = \frac{M_{\text{BH}}}{M}$$

For the Sombrero Galaxy:

$$\gamma_{\text{BH}}^{\text{Sombrero}} = \frac{10^9 M_{\odot}}{10^{11} M_{\odot}} = 0.01 = 1\%$$

2.2 SMBH Direct Gravitational Contribution

The BH contribution to g_{total} at the reference radius r :

$$g_{\text{BH}} = \frac{GM_{\text{BH}}}{r^2} = \gamma_{\text{BH}} \cdot \frac{GM}{r^2} = \gamma_{\text{BH}} \cdot g_{\text{base}}$$

Substituting $\gamma_{\text{BH}} = 0.01$ and $g_{\text{base}} = 2.382 \times 10^{-10} \text{ m/s}^2$:

$$g_{\text{BH}} = 0.01 \times 2.382 \times 10^{-10} = 2.382 \times 10^{-12} \text{ m/s}^2$$

2.3 UQFF Sphere of Influence

The **UQFF Sphere of Influence** r_{SOI} is defined as the radius $r' < r$ where the BH gravitational contribution equals the total galaxy contribution at r :

$$g_{\text{BH}}(r') = g_{\text{base}}(r)$$

$$\frac{GM_{\text{BH}}}{r'^2} = \frac{GM}{r^2}$$

Solving:

$$r'^2 = r^2 \cdot \frac{M_{\text{BH}}}{M} = r^2 \cdot \gamma_{\text{BH}}$$

$$r_{\text{SOI}} = r \cdot \sqrt{\gamma_{\text{BH}}}$$

For Sombrero:

$$r_{\text{SOI}} = 2.36 \times 10^{20} \cdot \sqrt{0.01} = 2.36 \times 10^{20} \times 0.1 = 2.36 \times 10^{19} \text{ m}$$

Physical interpretation: Within $r_{\text{SOI}} = 2.36 \times 10^{19} \text{ m}$ ($\sim 2.49 \text{ kly}$), the direct BH gravitational acceleration exceeds the galaxy-mean g_{base} . This is the UQFF-predicted boundary of BH gravitational dominance.

2.4 Verification

At $r' = r_{\text{SOI}}$:

$$g_{\text{BH}}(r_{\text{SOI}}) = \frac{GM_{\text{BH}}}{r_{\text{SOI}}^2} = \frac{G \cdot 0.01M}{(0.1r)^2} = \frac{0.01 \cdot GM}{0.01 \cdot r^2} = \frac{GM}{r^2} = g_{\text{base}} \checkmark$$

3. Comparative SMBH Dominance Table

Galaxy	M _{BH} (M _{sun})	M _{total} (M _{sun})	γ_{BH}	r _{SOI} / r _{ref}
Milky Way (Sgr A*)	$\sim 4 \times 10^6$	$\sim 10^{11}$	$\sim 4 \times 10^{-5}$	$\sim 6.3 \times 10^{-3}$
Andromeda (M31)	$\sim 1.4 \times 10^8$	$\sim 10^{12}$	$\sim 1.4 \times 10^{-4}$	$\sim 1.2 \times 10^{-2}$
M87	$\sim 6.5 \times 10^9$	$\sim 6 \times 10^{12}$	$\sim 1.1 \times 10^{-3}$	$\sim 3.3 \times 10^{-2}$
Sombrero (M104)	$\sim 10^9$	$\sim 10^{11}$	0.01	0.1

Sombrero's $\gamma_{\text{BH}} = 0.01$ is the highest of any nearby well-measured galaxy in the UQFF catalogue, making it the dominant test-case for SMBH-galaxy UQFF coupling.

Key comparison ratios: - Sombrero/Sgr A*: γ_{BH} ratio = $0.01/4 \times 10^{-5} = \mathbf{250\times}$ - Sombrero/M87: γ_{BH} ratio = $0.01/1.1 \times 10^{-3} \approx \mathbf{9\times}$ - Sombrero/Andromeda: γ_{BH} ratio = $0.01/1.4 \times 10^{-4} \approx \mathbf{71\times}$

4. UQFF BH Contribution in the Master Equation

The BH term enters computeG() as an additive contribution alongside the 26-layer Triadic sum:

$$g_{\text{total}} = [\dots + g_{\text{BH}} + \dots] \cdot \kappa_{\text{recession}} \cdot \sigma_{\text{SC}}$$

$$= [\dots + 2.382 \times 10^{-12} \text{ m/s}^2 + \dots] \times 0.99374 \times (1 - 10^{-20})$$

BH fractional contribution to g_{total} (estimated):

$$\frac{g_{\text{BH}}}{g_{\text{total}}} \approx \frac{2.382 \times 10^{-12}}{1.238 \times 10^{-8} + 2.382 \times 10^{-12} + \dots} \approx \frac{2.382 \times 10^{-12}}{1.24 \times 10^{-8}} \approx 1.9 \times 10^{-4}$$

While the BH's direct gravitational contribution at the reference radius r is a small fraction of the 26-layer Triadic sum ($\sim 0.019\%$), the r_{SOI} formula reveals that inside 2.36×10^{19} m, BH gravity dominates the reference-point baseline — a qualitative distinction for UQFF predictions of inner galactic structure.

5. Module Implementation

```
// PAPER_279 – SOMBRERO_UQFF_MODULE.cpp, updateCache()
gamma_BH = M_BH / M; // 0.01 = 1%
r_SOI = r * std::sqrt(gamma_BH); // 2.36e20 × 0.1 = 2.36e19 m

// Applied in computeG():
double g_BH = G_grav * M_BH / (r * r); // = gamma_BH * g_base = 2.382e-12 m/s2
```

Computed values for Sombrero M104:

Quantity	Value	Units
$M_{\text{BH}} = 10^9 M_{\text{sun}}$	1.989×10^{39}	kg
$M = 10^{11} M_{\text{sun}}$	1.989×10^{41}	kg
$\gamma_{\text{BH}} = M_{\text{BH}}/M$	0.01	dimensionless
$g_{\text{BH}} = G \cdot M_{\text{BH}}/r^2$	2.382×10^{-12}	m/s ²
$r_{\text{SOI}} = r \cdot \sqrt{\gamma_{\text{BH}}}$	2.36×10^{19}	m
r_{SOI} in kly	~2.49	kly

6. Key Constants Introduced

Symbol	Value	Units	Description
γ_{BH}	0.01	dimensionless	SMBH Dominance Ratio M_{BH}/M
r_{SOI}	2.36×10^{19}	m	UQFF Sphere of Influence radius
g_{BH}	2.382×10^{-12}	m/s ²	BH direct gravitational contribution at r

7. Physical Significance

- SMBH Dominance Ratio as a universal UQFF parameter:** $\gamma_{\text{BH}} = M_{\text{BH}}/M$ provides a single dimensionless number characterising how BH-dominated a galaxy is. It ranges from $\sim 10^{-5}$ (Sgr A*) to ~ 0.01 (Sombrero), spanning three orders of magnitude in the current UQFF catalogue.
- UQFF Sphere of Influence formula:** $r_{\text{SOI}} = r \cdot \sqrt{\gamma_{\text{BH}}}$ is derived directly from setting $g_{\text{BH}}(r') = g_{\text{base}}(r)$. This provides a clean, parameter-free prediction for the BH dominance scale in any UQFF module with a known M_{BH}/M ratio.
- Sombrero as extreme BH-galaxy system:** With γ_{BH} 250× larger than the Milky Way's, Sombrero tests UQFF behaviour in the BH-dominated regime. The large γ_{BH} implies that inner BH effects begin influencing the 26-layer Triadic structure at radii as large as $r_{\text{SOI}} = 2.36 \times 10^{19}$ m.
- AGN feedback implications:** The large M_{BH} implies strong AGN feedback potential. UQFF predicts that AGN activity modifies the vacuum energy density locally, which would appear in the UQFF framework as a time-varying component of $U_{\text{g1_vec}}[i]$ for the innermost layers — a direction for future UQFF module development.

5. **Generalisation:** The formula $\gamma_{\text{BH}} = M_{\text{BH}}/M$ and $r_{\text{SOI}} = r \cdot \sqrt{\gamma_{\text{BH}}}$ define a universal UQFF BH dominance prescription applicable to any galaxy module. Future UQFF modules for BH-dominated systems (e.g., NGC 1277, M87) should include this pair of parameters as standard.
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8. References

- Ford, H. C. et al. (1996). ApJ, 458, 132. (Sombrero M_{BH} from gas kinematics, HST)
 - Kormendy, J. et al. (1996). ApJ, 459, L57. (Sombrero BH mass confirmation)
 - Jarrel, J. R. et al. (2011). ApJ, 739, 21. (Sombrero DM halo and BH mass)
 - Gültekin, K. et al. (2009). ApJ, 698, 198. ($M_{\text{BH}}-\sigma$ correlation)
 - PAPER_277: UQFF Gravitational Recession Damping $\kappa_{\text{recession}}$ for $z = +0.0063$
 - PAPER_278: Sombrero Dust Ring UQFF Gravitational Ring Resonator ω_{ring}
 - SOMBRERO_UQFF_MODULE.cpp (UQFF 2.0, Session 77)
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UQFF 2.0 — The SMBH Dominance Ratio $\gamma_{\text{BH}} = M_{\text{BH}}/M$ and UQFF Sphere of Influence $r_{\text{SOI}} = r \cdot \sqrt{\gamma_{\text{BH}}}$ are new universal parameters for UQFF galaxy modules, first derived and tested on the Sombrero Galaxy. — Daniel T. Murphy, Session 77, March 2026.